Comprehensive Formula Reference: Indefinite Integral

Advanced Calculus & Competitive Examination Quick Reference

Prepared for High School, Senior Secondary (CBSE/ISC), and Engineering Aspirants

1. Basic Algebraic, Reciprocal, and Exponential Integrals

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + c$$

$$\int a^x\,dx = \frac{a^x}{\ln a} + c \quad (a > 0,\ a \neq 1)$$

$$\int e^{mx}\,dx = \frac{e^{mx}}{m} + c$$

$$\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \quad (n \neq -1)$$

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + c$$

2. Trigonometric Integrals (Standard Form using $\operatorname{cosec} x$)

$$\int \sin x\,dx = -\cos x + c$$

$$\int \cos x\,dx = \sin x + c$$

$$\int \tan x\,dx = \ln|\sec x| + c = -\ln|\cos x| + c$$

$$\int \cot x\,dx = \ln|\sin x| + c$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + c = \ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right| + c$$

$$\int \operatorname{cosec} x\,dx = \ln|\operatorname{cosec} x - \cot x| + c =$$

$$\int \sec^2 x\,dx = \tan x + c$$

$$\int \operatorname{cosec}^2 x\,dx = -\cot x + c$$

$$\int \sec x \tan x\,dx = \sec x + c$$

$$\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + c$$

3. Standard Rational and Inverse Trigonometric Integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \frac{-1}{\sqrt{a^2 - x^2}}\,dx = \cos^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \frac{1}{a^2 + x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \frac{1}{x\sqrt{x^2 - a^2}}\,dx = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + c$$

$$\int \frac{1}{a^2 - x^2}\,dx = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + c$$

$$\int \frac{1}{x^2 - a^2}\,dx = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + c$$

4. Standard Integrals Involving Quadratic Radicals

$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \ln\left|x+\sqrt{x^2+a^2}\right| + c$$

$$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \ln\left|x+\sqrt{x^2-a^2}\right| + c$$

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \sqrt{a^2+x^2}\,dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\ln\left|x+\sqrt{a^2+x^2}\right| + c$$

$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + c$$

5. Integration by Parts and Special Exponential Forms

The fundamental product integration identity:

$$\int u \cdot v\,dx = u\int v\,dx - \int\left(\frac{du}{dx}\cdot\int v\,dx\right)dx$$

Priority Rule for Choosing u (LIATE Sequence):

L (Logarithmic) $\longrightarrow$ **I** (Inverse Trigonometric) $\longrightarrow$ **A** (Algebraic) $\longrightarrow$ **T** (Trigonometric) $\longrightarrow$ **E** (Exponential)

Classic Exponential-Trigonometric & Linear Forms:

$$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}}{a^2+b^2}\big(a\sin(bx) - b\cos(bx)\big) + c$$

$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}}{a^2+b^2}\big(a\cos(bx) + b\sin(bx)\big) + c$$

$$\int e^x\big[f(x)+f'(x)\big]\,dx = e^x f(x) + c$$